AI Consumable YANG Data Model Discussion

Qin Wu, Kent Wasten (Moderators)

IETF ACTION Side Meeting

Wednesday 22 July 2026

14:45 - 15:45 (Vienna Time)

Park Suite 4

Meeting Agenda

1. Opening Remarks(IETF meeting rules, agenda arrangement, and objective) - 10 min Moderators
2. AI consumable YANG data model for IOT - 10 min + 5 min Oscar Eduardo LOPEZ
3. Rethinking YANG-based Service and Network management in the Era of Agentic AI - 15 min + 5 min Qiufang Ma
4. Open Discussion 15 mins

Background and Recap

- Model-driven programmability using YANG has become the norm
  - YANG becomes an important IETF legacy asset with 20+ year investment
    - https://www.ietf.org/blog/yang-really-takes-industry/
  - NEMOPS workshop revisited the network automation journey and identify the operation barriers for technology adoption [RFC9968].
- In the meanwhile AI protocols standardization discussion has become another hype
  - E.g., More work on AI for Network Operation has been proposed and distributed
  - AINETOPS ML has been created to help coordinate AI work
  - AI Driven Network Operation wiki page has been created to provide guidance
- It seems that YANG journey and AI journey take different courses

Why are we Here

- YANG is originally designed for network management, rigid, machine-centric.
- Insufficient for AI system such as Agentic workflow and LLM.
- Key Characteristics of data model structure suitable for AI,e.g., AI Agent, MCP Server
  - Ontology - Entities
  - Semantics – Properties of Entities
  - Constraints – Relation with Entities, Enumeration, Value Range
  - Context – Relation between Entities

Data Model structure options for AI

Option 1: ReUsing YANG to carry more rich semantics

- E.g., Symptom semantics defined in draft-ietf-nmop-network-anomaly-semantics

Option 2: Define new YANG model language extension (such as YANG 2.0)

Option 3: Translate YANG modelled data into Knowledge Graph data using RDF or other representation format

Option 4: Common Ontology vocabulary and Semantics Framework/Ontology modeling/Semantics Information Model

Relevant drafts for AI integration

Categories	Title	Relevant I-Ds
Option 1 Reuse YANG	Semantic Metadata Annotation for Network Anomaly Detection	draft-ietf-nmop-network-anomaly-semantics
Option 2 YANG Extension	**The YANG 2.0 Data Modeling Language**	draft-ietf-netmod-yang2-00
	Modelling Boundaries	draft-davis-netmod-modelling-boundaries-03
Option 3 YANG translation	Translation of YANG to other representation	See presentation in this meeting
	Guidelines for Translation of UML Information Model to YANG Data Model	draft-mansfield-netmod-uml-to-yang
	NAIM: A Canonical Data Format for AI-Assisted YANG Modeling	draft-feng-netmod-naim-00 draft-feng-netconf-naim-op
	Framework for Normalizing Multi-Vendor Network Inputs for LLM-Assisted Network Management	draft-prabhu-nmrg-prompt-schema-llm
Option 4 Ontology	Operational Requirements for Network State Exchange in Agent-Assisted Network Operations	draft-cui-nmop-agent-sketch-com-00
	Knowledge Graph Framework for Network Operations	draft-mackey-nmop-kg-for-netops

Questions/Comments/Concerns